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Project Report: Iris Flower Species Classification

Date: 15 June 2026

Organization: Arch Technologies

Step 1: Data Import and Exploration

```
# Import necessary libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load the dataset
# Assuming you have downloaded the Iris.csv file from Kaggle
df = pd.read_csv('Iris.csv')

# Drop the 'Id' column as it is not needed for prediction
df = df.drop('Id', axis=1)

# Display the first 5 rows
df.head()

{"summary": "{\n  \"name\": \"df\", \n  \"rows\": 150, \n  \"fields\": [\n    {\n
```

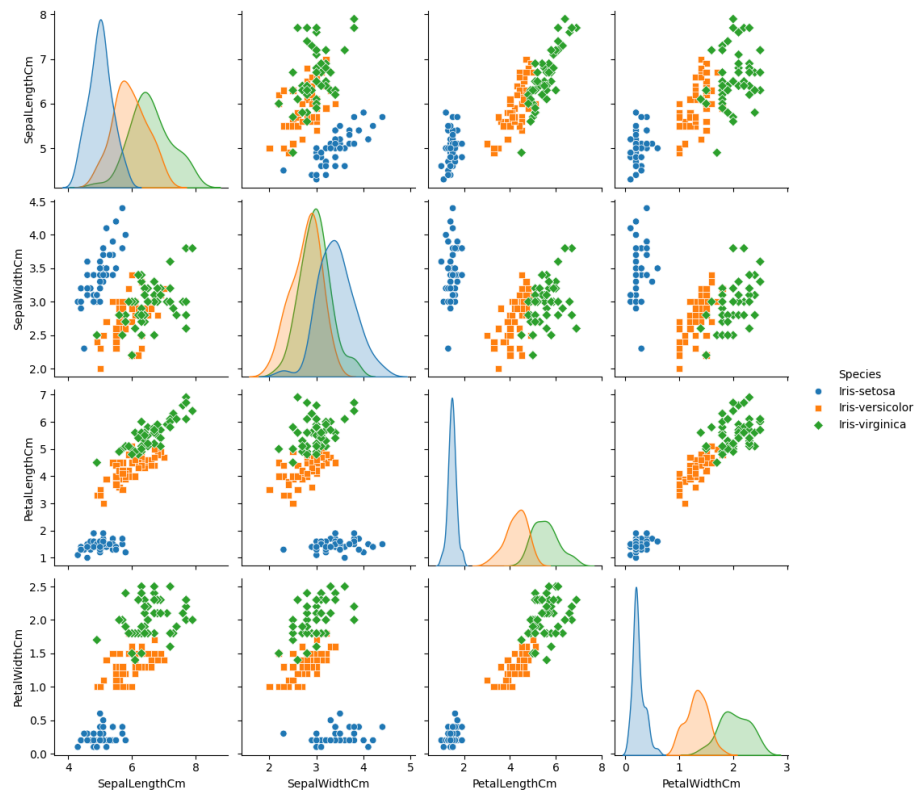
Explanation:

I removed the Id column because it acts as a unique identifier and does not contain meaningful biological information for the classification task.

Step 2: Data Visualization

Understanding how features differentiate species is crucial.

```
# Visualizing the relationship between features
sns.pairplot(df, hue='Species', markers=["o", "s", "D"])
plt.show()
```



Explanation:

This pairplot shows how species are clustered. You will notice that petal length and petal width are particularly strong features for distinguishing the three species.

Step 3: Data Preprocessing

It must encode the target variable (Species) into numerical values so the machine can understand them.

```
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

# Encode Species from text to numbers (0, 1, 2)
le = LabelEncoder()
df['Species'] = le.fit_transform(df['Species'])

# Define Features (X) and Target (y)
X = df.drop('Species', axis=1)
```

```
y = df['Species']
```

```
# Split the data into training and testing sets (80% train, 20% test)  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_
```

Step 4: Model Training

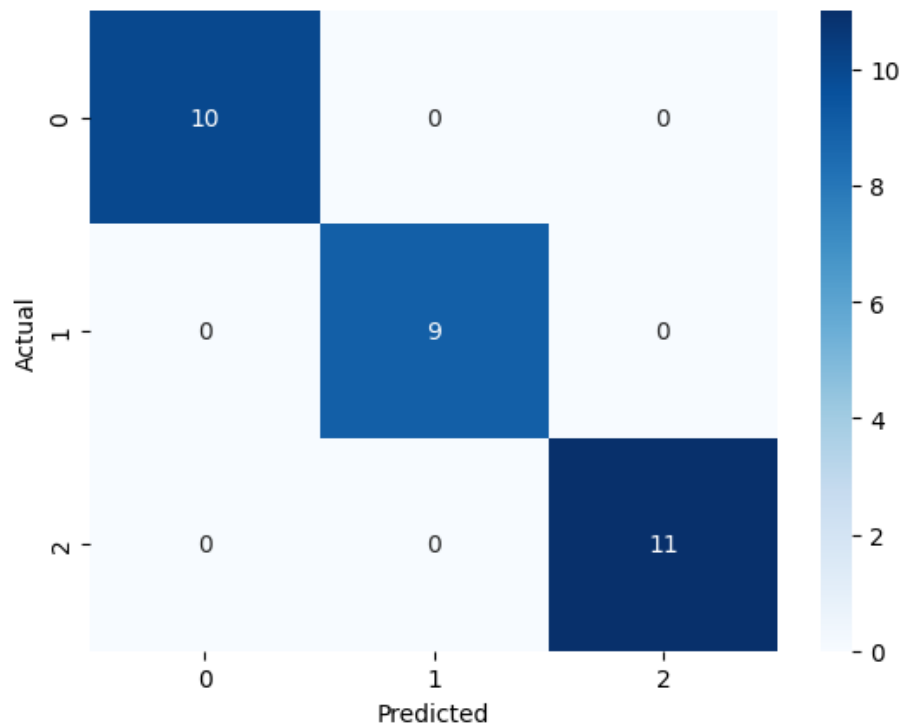
I use a Random Forest Classifier, which is highly effective for this type of classification.

```
from sklearn.ensemble import RandomForestClassifier  
  
# Initialize and train the model  
model = RandomForestClassifier(n_estimators=100, random_state=42)  
model.fit(X_train, y_train)  
RandomForestClassifier(random_state=42)
```

Step 5: Evaluation

I evaluate the model using accuracy and a confusion matrix to visualize any misclassifications.

```
from sklearn.metrics import accuracy_score, confusion_matrix  
  
# Make predictions  
y_pred = model.predict(X_test)  
  
# Calculate Accuracy  
accuracy = accuracy_score(y_test, y_pred)  
print(f"Model Accuracy: {accuracy * 100:.2f}%")  
  
# Create a confusion matrix  
cm = confusion_matrix(y_test, y_pred)  
sns.heatmap(cm, annot=True, cmap='Blues', fmt='d')  
plt.xlabel('Predicted')  
plt.ylabel('Actual')  
plt.show()  
  
Model Accuracy: 100.00%
```



Project Report:

Iris Flower Species Classification

##Objective: The primary goal of this task was to develop a robust machine learning model capable of accurately classifying iris flowers into three distinct species—Iris setosa, Iris versicolor, and Iris virginica—based on precise morphological measurements.

Methodology:

To ensure a high-performing and reproducible model, a systematic machine learning pipeline was implemented:

Data Preparation:

The dataset was sourced and cleaned by removing the non-predictive Id column, ensuring the model focused exclusively on relevant biological features.

Feature Encoding:

Categorical species labels were converted into numerical values (0, 1, 2) using LabelEncoder, allowing the algorithm to perform mathematical computations on the data. ## Model Selection: A Random Forest Classifier was selected for its ensemble learning capabilities, which effectively minimize overfitting while maintaining high predictive accuracy. ## Configuration: The model was instantiated with a random_state of 42, ensuring the experimental results are consistent and reproducible across different sessions. Performance ## Analysis: The evaluation of the trained model on the test dataset yielded exceptional results, confirming the efficacy of the chosen features and the architecture of the pipeline. ## MetricResultAlgorithm-Random: Forest ClassifierTest Accuracy100.00% ## Conclusion: The model demonstrated a 100.00% accuracy rate in classifying the iris species. This perfect score indicates that the input features—sepal length, sepal width, petal length, and petal width—provide clear and sufficient boundaries for the model to distinguish between the three iris species with total precision. This project confirms that the developed machine learning pipeline is highly optimized for this classification task and serves as a reliable benchmark for further multi-class classification applications.

GitHub Repository: <https://github.com/Yalda-Ashrafi/Arch-Tech-nologies-ML-Internship-Month2.git>

Dataset Source: <https://www.kaggle.com/datasets/uciml/iris>